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FREE FIELD TOOL | QUICK START GUIDE

AS9100D Gap Assessment *Workbook*

A detailed guide to a clause-by-clause readiness check against AS9100D: how the workbook is built, how it differs from the ISO 9001 version, how to record findings honestly and how to read the results.

88	4-10	5	FREE
CHECKPOINTS	AS9100D CLAUSES	WORKBOOK SHEETS	TO USE

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01 What this workbook is

The AS9100D Gap Assessment Workbook is a free Excel field tool for checking how ready an organization is against AS9100 Rev D, clauses 4 to 10. AS9100D contains the full ISO 9001:2015 requirements plus the aviation, space and defense additions, so one workbook covers both layers. It is built for a structured conversation on the shop floor, not for a desk exercise.

Who it is for

- **Quality and operations managers** preparing for AS9100D certification, a surveillance audit or a customer audit.
- **Internal auditors and consultants** running readiness assessments for a client, a site or a single program.
- **Aerospace and defense suppliers** that need to see where aerospace-specific controls lag the ISO 9001 base.

What is inside

SHEET	WHAT IT DOES
Instructions	How to use the workbook, the status values, the source column, the edition note and one completed example row.
Summary	Client, site, auditor and date, plus automatic results by clause and by requirement source: counts, assessed % and conformity %.
Checklist	88 checkpoints across clauses 4 to 10: requirement (paraphrased), source, audit question, evidence to request, common findings, follow-up question and related IAQG documents, then your input columns.
Aerospace Reference	Key AS9100D terms (counterfeit part, critical items, key characteristic, product safety, special requirements) and the supporting IAQG 9100-series documents.
AI Use Guide	Where AI helps in an audit workflow, the guardrails to respect (including export-controlled data) and starter prompts.

Coverage by clause

4 Context	5 Leadership	6 Planning	7 Support	8 Operation	9 Performance evaluation	10 Improvement	Total
5	5	5	15	46	7	5	88

Clause 8 holds over half of the checkpoints because aerospace audits go deepest there. Each checkpoint also carries a source tag:

ISO 9001 base	AS9100D addition	Mixed (both in one row)
38	21	29

What you need

- The workbook in **Excel** (desktop version recommended; dropdowns and formulas work best there).
- A **licensed copy of AS9100D** (it includes ISO 9001:2015) to check any requirement you are unsure about.
- Access to the **process owners and their records** for your scope, so evidence can be seen, not described.

BEFORE YOU START

Clause text is **paraphrased for field use**. AS9100D is published by SAE International and must be obtained from SAE or your national standards body: verify requirements against the licensed standard. The workbook supports, and does not replace, auditor judgment. Findings are **indicative**, not a certification decision.

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02 ISO 9001 or AS9100D: which workbook?

I also publish a free ISO 9001:2015 Gap Assessment Workbook. The two tools share the same layout and way of working. They differ in depth and in who they are for.

	ISO 9001 WORKBOOK	AS9100D WORKBOOK
Standard	ISO 9001:2015	AS9100 Rev D (September 2016), which includes ISO 9001:2015
Checkpoints	36	88
Level of detail	Related sub-clauses grouped into one checkpoint	Sub-clause level; for example, clause 8.4 on external providers is split into 9 checkpoints
Checklist columns	Requirement, audit question, evidence, common findings, follow-up question	The same, plus **Source** (base or aerospace addition) and **Related IAQG docs**
Summary results	By clause	By clause and by requirement source
Sheets	4	5, adding the Aerospace Reference
Best for	Building or checking a sound quality system; organizations new to ISO 9001	Aviation, space and defense suppliers preparing for AS9100D, a customer audit or a check of their aerospace controls

Which one first? If the quality system is not yet stable at ISO 9001 level, fix that first with the ISO 9001 workbook. If the organization already works to ISO 9001 and supplies aerospace, go straight to the AS9100D workbook: it covers the base and the additions in one pass, so you do not need to run both on the same scope.

Where AS9100D goes beyond ISO 9001

AREA	WHAT THE CHECKLIST TESTS	CLAUSES
Management representative	A named representative with authority and free access to top management	5.3
On-time delivery	Delivery measured and acted on in leadership, customer satisfaction and management review	5.1.2, 9.1.2, 9.3.2
Measuring equipment	Register with the required fields, recall process, personal and customer-supplied equipment	7.1.5.2
Operational risk	A defined process to assess, communicate, mitigate and formally accept risk in operations	8.1.1
Configuration management	Product identity, traceability and agreement between documents and the actual product	8.1.2
Product safety	Hazard assessment, safety critical items, event reporting and training	8.1.3, 7.3
Counterfeit parts	Prevention, source control, detection, quarantine and reporting	8.1.4, 8.4.2, 8.7
Critical items, key characteristics	Identification and control from design to shipping	8.1, 8.3.5, 8.5.1
External providers, work transfer	Approval status register, delegated verification, test report validation, flow-down, right of access; planned transfer of work	8.4, 8.1
Production controls	Human error prevention, foreign object control, utilities, validation of equipment and software	8.5.1
Special processes, first article	Approval, qualification and revalidation; first article inspection and its repeat triggers	8.5.1.2, 8.5.1.3
Nonconformance, corrective action	Design-authority approval for use-as-is and repair, customer authorization, scrap control; human factors in root cause, flow-down to providers	8.7.1, 10.2.1

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03 Run an assessment in six steps

1	**Set up**	On the Summary sheet, enter the client, site or scope, auditor and assessment date. Decide the boundary first: one process, one department, one site or one program. Confirm that AS9100D is the right standard (maintenance organizations use 9110, distributors 9120) and which edition your customer and certification body apply. Mark design (8.3) not applicable only if design is outside the scope, and write the justification.
2	**Choose your trails**	Aerospace audits follow product, not clauses. Before walking the floor, pick a recent part number, a serial number, a key characteristic, an external provider, a first article and a nonconformance. Section 05 shows how to use them.
3	**Assess**	On the Checklist sheet, work clause by clause. Ask the audit question, request the evidence listed, compare what you see with the common findings and use the follow-up question to test how deep the system goes.
4	**Record**	Fill the shaded cream cells only: Status (dropdown), Evidence seen / notes, Owner and Target date. Everything else is reference content or formulas.
5	**Review**	The Summary sheet updates automatically: counts per clause and per source, assessed % and conformity %. Read the assessed % first.
6	**Act**	Turn every gap into an action with an owner and a date. Re-assess the same clauses after the actions are closed to confirm the gap is really gone.

The five status values

STATUS	MEANING
Conforming	The requirement is met and you saw the evidence yourself.
Minor gap	The requirement is partly met, or the lapse is isolated. The system still works overall.
Major gap	The requirement is missing, or the lapse is systemic or puts conformity of product at risk.
Not applicable	The requirement does not apply to the scope. Justify it in the notes (for example, design excluded from the scope under 8.3).
Not assessed	Default value. Not yet checked, so it is left out of the percentage until you assess it.

These descriptions are working guidance. The final classification of any finding remains the auditor's judgment, and your certification body or customer may apply its own criteria.

> **TIP FOR THE CHECKLIST SHEET**
>
> Use the **Common findings** column as a search list, not a script. If the organization is clean on the usual traps, push further with the **Follow-up question**. The most useful notes come from asking to see a record or to observe an operation, not from asking whether a procedure exists. Use the **Source** filter to look at the AS9100D additions on their own.

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04 Read the results and score honestly

The Summary sheet shows, for each clause and for each requirement source, how many items are conforming, minor gap, major gap, not applicable and not assessed, with a conformity percentage. It also gives the total assessed % and the number of open gaps (minor plus major).

> **HOW CONFORMITY % IS CALCULATED**
> Conformity % = Conforming ÷ (Items - Not applicable - Not assessed). Items with no status yet are excluded until you assess them. A high conformity % means little if the Assessed % is low, so read the assessed % first. The result is an indicative readiness measure, not a certification outcome.

Reading the two tables together

- **Good base, weak additions.** A healthy ISO 9001 base result with a low AS9100D addition result points to aerospace-specific controls (risk, configuration, product safety, counterfeit parts, special processes, first article) as the first place to act.
- **Weak base.** Gaps in the ISO 9001 base rows, such as documents, audits or corrective action, will undermine the additions. Fix the foundation first. In mixed rows, check whether the gap sits in the base or the aerospace part.

Scoring honestly

- **Score what you saw, not what is written.** A procedure on file is not evidence that it is followed.
- **No evidence, no “Conforming”.** If you could not see the record or observe the activity, use Minor gap or leave it as Not assessed.
- **Sample independently.** Pick the records yourself (five gauges, three purchase orders, two first articles) instead of accepting the ones offered.
- **A certificate is not control.** A provider certificate or test report on file is an input. Check scope, expiry, revision and whether anyone reviewed it.
- **Do not use “Not applicable” to skip hard clauses.** Every use needs a written justification tied to the scope.
- **Write specific notes:** what you checked, how many, which record, what was missing. Another person should be able to repeat your check.

A completed row

CLAUSE	**8.4.1.1** Provider approval status and register
STATUS	Minor gap
EVIDENCE SEEN / NOTES	Approved provider register reviewed; 4 of 60 active providers have no approval scope recorded. 1 of 5 sampled purchase orders (PO 1042) placed with a provider marked conditional, no documented release for the conditional use. Customer-designated source list not cross-checked against the register.
OWNER	Purchasing Manager
TARGET DATE	2026-12-15

Why this row works: it names the records reviewed, gives counts and references, states the gap in terms of evidence, and gives a named owner and a date.

Common mistakes

- Assessing every clause in one sitting with a single person. Involve the process owners for 7, 8 and 9.
- Reading the Summary as a pass or fail result. It points to where to look first; it does not decide certification.
- Closing gaps on paper only. Re-check the evidence after the corrective action is complete.

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05 Five trails that show if the system is real

Aerospace audits are trail-based. Instead of checking one clause at a time, pick one real item and follow it through the system. If the trail breaks, the checklist tells you which clauses to score. These five trails cover most of the AS9100D additions.

TRAIL	WHAT TO FOLLOW	CLAUSES
1 Serial number	From a shipped serial number back to material, process records, inspection and configuration, then forward from the material batch to every destination.	8.5.2, 8.1.2, 8.6
2 Key characteristic	From the drawing flag to the traveler, inspection plan, measurement data and capability evidence, and to the operator who has to control it.	8.1 f, 8.5.1, 8.3.5
3 External provider	From one purchase order to the provider register, approval scope, flow-down clauses, receiving verification and any delegation.	8.4.1.1, 8.4.2, 8.4.3
4 First article	From a new or changed part to its first article inspection, and from an engineering, tooling or process change to the repeat inspection it should trigger.	8.5.1.3, 8.5.6, 8.3.6
5 Nonconformance	From a use-as-is or repair disposition to the design authority approval, the customer authorization, the correction and the re-verification.	8.7.1, 10.2.1

How to use a trail

- Pick the item yourself, ideally from the last three months, and ask for every record along the way.
- At each step, note the clause on the Checklist sheet and score it with what you saw.
- Break in the trail = gap. Record where it broke, what was missing and who owns the step.
- Repeat with a second item. One clean trail can be luck; two are a pattern.

> **WHY TRAILS WORK**
>
> A system can look complete clause by clause and still fail at the joints: a key characteristic that never reaches the inspection plan, a provider that is used outside its approved scope, a change that never triggers a new first article. Trails find the joints.

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06 Using AI safely in an audit workflow

The workbook includes an AI Use Guide sheet. AI can save preparation time, but only if you keep confidential data out of it and keep the judgment with the auditor. In aerospace and defense, confidentiality includes export-controlled technical data.

WHERE AI HELPS	GUARDRAILS
• Draft questions tailored to a process or product. • Evidence lists built from a clause, to send before the audit. • Finding wording drafted from your raw notes. You verify every word. • Summaries of long procedures or earlier reports.	**Confidentiality.** Do not enter client names, part numbers, drawings, contract terms, customer proprietary data or export-controlled technical data (for example ITAR or EAR). Sanitize first (“Client A”, “Part X”). **Verification.** AI can misstate requirements, and AS9100D differs from ISO 9001 in many small ways. Check every clause reference against the licensed standard. **Judgment.** Classifying findings and drawing conclusions stays with the auditor. **Record.** Note in the audit file where AI assisted the drafting.

STARTER PROMPT: EVIDENCE REQUEST

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“Act as an AS9100D auditor. For clause 8.4.1.1, list the documents and records I should request from a small machining supplier about its external provider approval register. Use generic placeholders; no client data.”
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07 Next steps

- **Take the free AS9100D readiness test:** cristianlabella.com/as9100d.html
- **Download the ISO 9001 workbook:** cristianlabella.com/downloads/ISO9001_2015_Gap_Assessment_v1.0_Protected.xlsx
- **Take the free Manufacturing Readiness Assessment** (24 questions, about 5 minutes): cristianlabella.com/assessment.html
- **Share your feedback.** If you use the workbook on a real assessment, I would value your comments on what worked and what was unclear.

EDITION NOTE

This workbook is based on **AS9100 Rev D (2016)**. The IAQG is revising the 9100 series, to be published as IA9100. Check which edition your customer and certification body apply. An updated version of this workbook is planned once the new edition is published.

ABOUT THE AUTHOR

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